Prism Market

Technical whitepaper

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Version History

Version	Date	Description
1.0	February 21st 2026	Release of the Prism Market technical whitepaper.

Abstract

Decentralized prediction marketplaces have evolved rapidly from early experiments in information aggregation to robust platforms leveraging decentralised technologies. There are numerous centralised prediction marketplaces currently operating on the Internet - all of which are under centralized control and a high level of trust in the marketplace operator is needed for users to confidently participate. Decentralized ledger technology enables trustless participation in prediction markets, censorship resistance, much lower transaction costs and reduced reliance on intermediaries. As distributed ledger technology matures, decentralized prediction markets are poised to unlock new forms of collective intelligence, democratize access to forecasting tools, and serve as foundational primitives for risk management and decision-making.

Prism Market is a decentralized prediction market and prediction protocol. Prism provides a high-speed, low-latency, low-trust, transparent, and orderly marketplace for forecasting outcomes of any verifiable statement.

Decentralized prediction marketplaces enable users to trade positions on some market statement in a peer-to-peer way [3]. Any user can create a new market (a market statement) and any user can open or close a position in that market by interacting with a decentralized state machine such as a smart contract [8]. Because the Prism marketplace is decentralized, users can exchange predictions with each other in a peer-to-peer way, which is in contrast to relying on a centralized platform to facilitate order matches [19].

Prism utilises a hybrid architecture where high-frequency order matching is performed off-chain and order settlement is finalised on-chain. Hybrid execution architectures combining off-chain execution with on-chain settlement are widely studied as a scalability strategy for distributed ledgers [9]. A highly scalable, low-latency off-chain central limit order book (CLOB) is used to match signed order intents. Matched order intents are then settled on-chain using a smart contract (e.g. EVM) [18]. This hybrid architecture provides a balance between users' needs for low latency and high throughput with users' limited trust of centralized platforms and a preference for ultra-low transaction fee, uncensorable, peer-to-peer value transfer mechanisms [6].

Prism is built on the Hedera network which features massive transaction-per-second throughput, low latency, security guarantees, ordering fairness, low and predictable transaction fees [10][14]. Additionally, the Hedera network is governed by a governing council which ensures long-term stability [10] of the network. The technical and governance features of the Hedera network make it particularly suitable for building applications such as prediction marketplaces, which have the potential to grow to many millions of users and many millions of markets. Prediction marketplaces are just one category of application that could run on a trust network, so choosing to build on a network such as Hedera that can grow and scale into the future is an intentional decision.

This technical whitepaper details the implementation of Prism, including its central limit order book (CLOB), its signature-based on-chain settlement mechanism, describes the capability of the Hedera base layer, and discusses how the protocol can evolve into the future with further decentralization and performance upgrades.

1 Introduction

Prism is a decentralized marketplace which facilitates trading on the outcome of future events. Any account (human or agent) can create a market by declaring a market statement and submitting the transaction to the Prism smart contract. Any account (human or agent) can buy a position in the market by submitting an order intent to the Prism smart contract with an accompanying collateral token (Circle USDC is Prism's collateral token). When the market resolves, the position settles and the winning position is paid out to the account using the collateral token. A losing position forfeits its collateral in its entirety.

Prediction markets have been shown to aggregate distributed information efficiently and often outperform traditional forecasting methods [1][2][3]

At the protocol level, any participant – whether human or algorithmic agent – can propose a new event via the **Market Factory**. The Prism protocol is a decentralised protocol and as long as the outcome of a statement can be readily verified by a decentralized resolution source (**Oracle**), the Factory will proceed to create a new market. Once the market is live, participants can buy a position by submitting a cryptographically signed order intent to the Prism smart contract, funded by USDC collateral. Upon market resolution, winning positions settle on-chain and are paid out in USDC, while losing positions forfeit their entire collateral.

1.1 User Journey

A typical user journey through `prism.market's` hybrid architecture proceeds as follows:

- Market creation
- Order intent creation and signing
- Order submission
- Order match
- Order settlement (on-chain)
- Market settlement
- Collateral token redemption

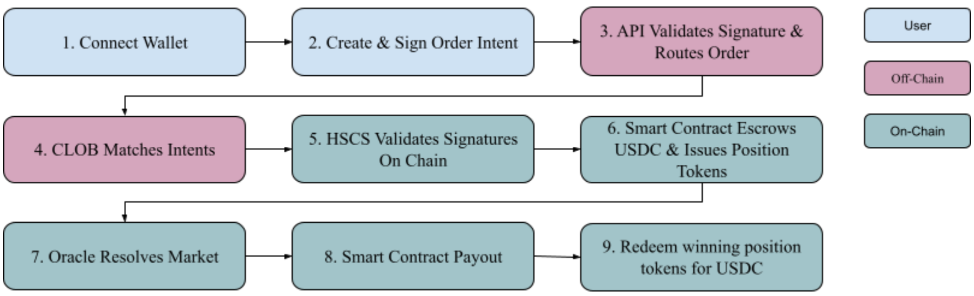

Figure 1: Atomic Order Matching Flow Diagram

The prediction market logic in figure 1 was implemented for the Ethereum virtual machine. The EVM can process a high number of transactions per second and can process reasonably large instructions of byte code without incurring too many fees (measured in USD). The low cost execution of the EVM running on the Hedera network makes low value predictions feasible – such predictions would not be feasible on other networks. Additionally, there are no layer 1 or layer 2 methods on Hedera which many gen 1 and gen 2 blockchains (e.g. bitcoin and ethereum) utilise to try to address scalability challenges. The major problem with such an approach is the reduction in decentralisation and security tradeoffs which actually lowers trust and increases centralised control in the market. The bigger the scalability problems, the worse this situation becomes and there can even be a cannibalisation effect on the base layer where transaction validators aren't incentivised to process transactions because the transaction flow is being redirected towards the higher scalability layer(s).

The prism smart contract makes extensive use of the Hedera pre-compiled smart contracts which are heavily tested, verified and execute with a high degree of reliability and gas efficiency. The prism smart contract performs on-chain signature verification of all order intents and supports multiple cryptographic key types such as ECDSA and ed25519.

Algorithm 1 Atomic Order Matching

Require: YES orderIntent, NO orderIntent

```
if market is resolved or not initialized then
    abort
end if
Reject duplicate txIds which have already settled in full
Compute collateral and quantity for YES and NO orderIntents
Determine minimum collateral and minimum quantity (partial match)
Cryptographically verify signatures for both orderIntents
Transfer minimum collateral from both participants to smart contract
Update market collateral total
Assign minimum quantity of YES tokens to YES buyer and NO tokens to NO buyer
Mark orderIntents as partially or fully matched
Emit events and return updated balances
```

2 Prism Architecture

Prism is built on Hedera which features high transaction throughput, scalability, transaction ordering fairness, low latency and predictable fees [10][14]. Importantly, the Hedera Council which governs the network ensures its long-term stability [10], orderly, predictable operation and enables organisations to forecast their operating costs with clarity. Prism is a Hedera-only protocol and inherits all the benefits and trust that is inherent to the Hedera network.

From a high level architecture perspective, the Prism protocol consists of two core components:

1. **On-chain smart contract settlement mechanism** (decentralized)
2. **Off-chain matching of signed order intents** (semi-centralized)

Any person or entity with a Hedera account can create a prediction intent, sign it with their Hedera wallet, and submit it to the Prism API where it is validated before being routed to the Prism CLOB [17].

The Prism CLOB is a distributed, high-performance order matching engine that processes thousands of transactions per second and can scale horizontally using standard cloud-based approaches (load balancing, container replication, Kubernetes, etc.).

The Prism smart contract is written in Solidity and is published on the Hedera network. The smart contract is notified of order intent matches by the Prism API. The smart contract validates the match on-chain (including digital signature verification) and performs collateral token transfer as well as position token allocation. The smart contract design prioritises safety and features the following security precautions [18][16]:

- Tested smart contract
- Audited smart contract
- Hedera pre-compiled smart contract interfaces
- Testnet bug bounty programme
- Battle-tested on testnet

2.1 Off-Chain Matching Engine

To provide a high-quality trading experience on a par with that experienced on centralized exchanges, Prism Market utilizes a high-performance off-chain matching engine. The matching engine is written in Rust which results in a CLOB that is performant, low-latency, memory safe, CPU/memory efficient, reliable and scalable.

The prism matching engine (CLOB Algorithm 2) processes incoming orderIntents for particular a prediction marketId and compares them against existing orders on the opposite side (YES side vs NO side). The CLOB matches orders based on price and quantity: a YES order matches if its price is greater than or equal to the absolute value of a NO order's price, and vice versa. If the incoming order's quantity is less than or equal to the existing order's quantity, a full match occurs; otherwise, a partial match is made and the remaining quantity is matched with other orders. Matched orders trigger notifications via an event bus. As the protocol evoles, in future, events from the Prism event bus may be made available to subscribers in real-time. Unmatched orders are added to the order book for future matching. The engine is designed for high performance and all incoming orderIntent objects undergo rigourous validation for syntactic and semantic correctness.

Matching orderIntent objects are forwarded to a smart contract for on-chain settlement. The Prism smart contract verifies signatures using standard cryptographic validation techniques [15][16].

Non-Custodial Architecture
User funds remain under cryptographic control until execution conditions are satisfied, a core principle of decentralized applications [8].

Immutable Settlement
Final settlement is immutable once confirmed, reflecting the tamper-resistant guarantees of distributed ledger consensus [14].

Low Latency
The Prism orderbook rapidly matches prediction intents in under a millisecond. Such order matching speed and latency would not be possible to perform on-chain.

Near Real-Time Responsiveness
From the moment a prediction intent is made given the current market information, to the moment an order is matched and sent to the Prism smart contract, a user experiences near-real-time responsiveness.

Orderbook Transparency
All pending orders are publicly visible to market participants via a real-time streaming gRPC interface.

Low Fees
By processing order intents off-chain, users do not incur Hedera network fees for placing or canceling unmatched orders.

Familiar Interface
The orderbook user interface is familiar to traders and DeFi users.

Algorithm 2 Order Matching Engine (High-Level Pseudocode)

```
Require: Incoming order, Order book (opposite side and same side)
Sort opposite side orders by price
for each order in opposite side orders do
    if price constraints are satisfied then
        if incoming order quantity ≤ existing order quantity then
            Perform full match
            Update quantities and remove matched order if depleted
            Notify match event (full) via NATS
            return
        else
            Perform partial match
            Reduce incoming order quantity, remove matched order
            Notify match event (partial) via NATS
            Continue matching with remaining quantity
        end if
    else
        Break (no further matches possible)
    end if
end for
Add unmatched incoming order to same side order book
```

2.2 On-Chain Settlement (HSCS)

User order intents are settled on-chain using the Hedera Smart Contract Service. Two matching order intent objects (including digital signatures) are presented to the smart contract for settlement. The smart contract validates the objects, verifies the digital signatures (using a Hedera pre-compiled smart contract), holds collateral tokens in decentralised escrow, and allocates redeemable position tokens. The contract verifies signatures using standard cryptographic validation techniques [15][16].

Non-Custodial
The Prism prediction protocol is non-custodial; user funds are held in smart contracts and can only be moved on a valid market determination by an oracle, valid trade execution, or user-initiated redemption.

Immutable Settlement
Final trade execution and the distribution of outcome shares are finalized on-chain, ensuring that once a match is settled, it is irreversible and uncensorable.

Non-custodial architecture ensures user funds remain under cryptographic control until execution conditions are satisfied, a core principle of decentralized applications [8]. Final settlement is immutable once confirmed, reflecting the tamper-resistant guarantees of distributed ledger consensus [14].

Ancillary functionality is encoded in the smart contract to handle smart contract instantiation, new market creation, set market outcomes, redeem position tokens for collateral tokens, retrieve balances, etc.

2.3 A generalized prediction protocol

A prediction intent is presented to prism.market using the rigidly defined format (protobuf) in listing 1. A human or an agent can construct a predictionIntentRequest object for any of the available markets, sign the prediction intent object using their Hedera wallet and present their prediction to prism.market for peer-to-peer matching and settlement. prism.market has made a public and permissionless gRPC interface (web) available and this is the primary means of access for any person or agent wishing to participate in a Prism prediction market.

Listing 1: PredictionIntentRequest Protobuf Definition

```
message PredictionIntentRequest {
  string tx_id = 1;
  string net = 2;
  string market_id = 3;
  string generated_at = 4;
  string account_id = 5;
  string market_limit = 6;
  double price_usd = 7;
  double qty = 8;
  string sig = 9;
  string public_key = 10;
  string evm_address = 11;
  uint32 key_type = 12;
}
```

Public interaction with the Prism protocol is supported through the highly-available Prism API. Direct interaction with the smart contract (e.g. redeem collateral tokens after successful prediction) is also provided for [15].

2.4 Auditable History (HCS)

Transparency is achieved through consensus-ordered event logs (Hedera Consensus Service — HCS). Append-only event streams provide verifiable audit trails analogous to distributed log systems [17]. Tamper-resistant logging enables independent verification of market history without relying on the matching engine operator. Real-time event streams allow automated agents to detect opportunities, similar to streaming market data infrastructure used in traditional finance systems [17].

Transparency and trust are enhanced using HCS. All critical protocol events, including market creation and finalized trades, are logged via HCS. The Prism event stream can be listened to by any market participant and the provenance of each matched order can be readily and publicly verified with no permissions required.

Tamper-Proof Audit Trail
: Prism's HCS event stream creates a verifiable record of all market activity that is independent of the off-chain matching engine.

Prediction Event Stream Broadcast
: Interested parties (including bots) can listen to Prism's event stream in real-time and have the ability to quickly identify verifiable market opportunities without delay or middle-man interference.

3 Market Mechanics

3.1 Binary Outcomes & Collateral

Prism is a binary option (YES/NO) prediction market. In the future, Prism may support outcomes other than YES/NO. Binary contracts are a standard structure in prediction market design literature [1].

Stablecoin collateral provides price stability relative to fiat currency and is commonly used in decentralized finance markets [13][19].

Collateral Token
Currently the only supported collateral token is Circle's USDC.

Position Token
The Prism smart contract converts collateral tokens (USDC) to position tokens according to the current market price and the amount of available liquidity.

All markets on Prism are fully backed by USDC collateral — there is no borrowing capability and all markets are fully collateralised. Fully collateralized prediction markets eliminate counterparty risk and margin insolvency concerns [1].

A market on the Prism smart contract has the following properties:

- A unique identifier (UUID)
- A market statement (text)
- A ledger which maintains USDC collateral balances (escrow) for a particular user
- A mechanism to convert USDC collateral into YES/NO position tokens according to the instantaneous market price
- A ledger which maintains YES/NO balances for a particular user

Share Value
Prices float between $0.00 and $1.00, representing the market's estimated probability (e.g., $0.65 = the market believes a YES outcome has a 65% probability and a NO outcome has a 35% probability). Prediction market prices are widely interpreted as probabilistic forecasts under rational market assumptions [3].

Settlement
Upon resolution of the binary option market, winning shares are redeemable for 1.00 USDC, while losing shares expire worthless.

Oracle Provider
Only an authorized oracle can resolve a market. Oracle-based resolution is a standard component of decentralized prediction protocols [8].

3.2 CLOB vs. AMM

Unlike AMMs that use bonding curves, Prism uses a Central Limit Order Book (CLOB) where prices are discovered through direct competition between buyers and sellers.

Efficiency
This model allows for tighter spreads and more accurate probability discovery.

No Impermanent Loss
Traders avoid the "impermanent loss" typical of AMM-based prediction markets.

3.3 Fee Structure & Revenue Model

Prism utilizes a revenue model designed to encourage maximum trading volume and user (human and agent) participation while ensuring long-term sustainability. Low initial fees are commonly used in platform bootstrapping strategies to stimulate network effects [3].

Genesis Interface Fees
Currently set to 0% to minimize friction for early adopters and bootstrap liquidity.

Redemption Fee
Protocol revenue is generated solely at settlement. A 2% fee is applied only when winning shares are redeemed for collateral.

Governance Flexibility
Fee parameters are adjustable by the protocol governance (Prism DAO) to ensure adaptability. Future adjustments may be implemented to fund development, insurance funds, or token buybacks, subject to community consensus.

4 Hedera Native Integration

Prism Market is engineered to maximize the efficiency of the Hedera network's native services.

HSCS
The protocol utilizes HSCS for trustless settlement logic. In future, the protocol may evolve to support Hedera Token Service (HTS) to provide native share management with higher speed, security, lower cost, and higher transaction-per-second throughput. Native token services can provide efficiency and throughput advantages over contract-level token implementations [12].

Transparency and Trust using HCS
Prism uses HCS topics to log every matching engine payload, ensuring the off-chain state can be publicly verified thereby increasing trust in the protocol and reassuring market participants that the market conducts itself in an orderly way that can be trusted and payouts are guaranteed within the limits of the protocol [11].

Predictable Economics
Leverages Hedera's fixed-fee model to ensure that market creation and resolution costs remain low and stable, regardless of network congestion [10].

5 Technical Specifications

Note: The following sections define the specific technical implementation of the architecture described above. These parameters are critical for developer and auditor reference.

5.1 Matching Engine Stack

Language & Architecture
Rust orderbook.

Latency Benchmarks
TBD. Latency will vary according to resource demands, memory size, and processing power. Additionally, network loading is an important factor which can affect latency of the CLOB.

5.2 Oracle Framework

Providers
TBD — UMA, Chainlink, etc.

Aggregation Logic
N/A — only one oracle provider for MVP.

5.3 HIP-206 Implementation

Precompile Usage
Pre-compiled Hedera smart contracts are used for on-chain digital signature verification.

5.4 HCS Schema

Topic Structure
Schema not currently defined. Messages are published consistently.

6 Disclaimer

This technical whitepaper is for general information purposes only. It does not constitute investment advice or a recommendation or solicitation to buy or sell any investment. It should not be relied upon for accounting, legal, tax advice, or investment recommendations.

References

[1] R. Hanson, *Combinatorial Information Market Design*, Journal of Prediction Markets, 2003.

[2] J. Berg, F. Nelson, T. Rietz, *Prediction Market Accuracy in the Long Run*, International Journal of Forecasting, 2008.

[3] J. Wolfers, E. Zitzewitz, *Prediction Markets*, Journal of Economic Perspectives, 2004.

[4] E. Budish, P. Cramton, J. Shim, *The High-Frequency Trading Arms Race*, Quarterly Journal of Economics, 2015.

[5] P. Daian et al., *Flash Boys 2.0*, IEEE Symposium on Security and Privacy, 2020.

[6] H. Adams et al., *Uniswap v3 Core Whitepaper*, 2021.

[7] V. Buterin, *A Next-Generation Smart Contract Platform*, 2014.

[8] J. Poon, T. Dryja, *The Lightning Network*, 2016.

[9] L. Baird, *The Hashgraph Consensus Algorithm*, 2016.

[10] Hedera, *Hedera Whitepaper*, Latest version.

[11] Hedera, *Consensus Service Documentation.*

[12] Hedera, *Token Service Documentation.*

[13] G. Wood, *Ethereum Yellow Paper.*

[14] N. Szabo, *Formalizing and Securing Relationships on Public Networks*, 1997.

[15] J. Katz, Y. Lindell, *Introduction to Modern Cryptography*, 2020.

[16] D. Boneh, V. Shoup, *Applied Cryptography*, 2020.

[17] J. Kreps, N. Narkhede, J. Rao, *Kafka: A Distributed Messaging System*, 2011.

[18] NASDAQ, *Matching Engine Architecture Overview*, 2019.

[19] R. Lyons, G. Viswanath-Natraj, *What Keeps Stablecoins Stable?*, NBER, 2020.